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J.J. College of Engineering and Technology

Department of Mechanical Engineering

ME25CO8

Metallurgy and Materials Science

Student Notes

Unit - IV: Non-Metallic Materials

Scheduled Hours: 9 Periods

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Year and Semester : 2nd Yr and III Sem

Course Handling Faculty

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Course Objectives

1. To learn the construction of phase diagrams and the use of the Iron–Iron carbide phase diagram for understanding microstructure formation.
2. To learn the selection and application of various heat treatment processes and their influence on microstructure formation.
3. To illustrate the different types of ferrous and non-ferrous alloys and their uses in the engineering field.
4. To illustrate different polymers, ceramics and composites and their uses in the engineering field.
5. To learn various testing procedures and failure mechanisms in engineering materials.

Full Course Content

Course Code: ME25CO8

Course Title: Metallurgy and Materials Science

Unit	Content
UNIT I	CONSTITUTION OF ALLOYS AND PHASE DIAGRAMS Constitution of alloys – solid solutions, substitutional and interstitial phase diagrams, isomorphous, eutectic, eutectoid, peritectic and peritectoid reactions, Iron–Iron carbide equilibrium diagram. Classification of steel and cast-iron microstructure, properties and application.
UNIT II	HEAT TREATMENT Definition – full annealing, stress relief, recrystallisation and spheroidising – normalizing, hardening and tempering of steel. Isothermal transformation diagrams – cooling curves superimposed on I.T. diagram – continuous cooling transformation (CCT) diagram – austempering, martempering – hardenability, Jominy end quench test – case hardening, carburizing, nitriding, cyaniding, carbonitriding – flame and induction hardening – vacuum and plasma hardening – thermo-mechanical treatments – elementary ideas on sintering.
UNIT III	FERROUS AND NON-FERROUS METALS Effect of alloying additions on steel (Mn, Si, Cr, Mo, Ni, V, Ti and W) – stainless and tool steels – HSLA – maraging steels – grey, white, malleable, spheroidal alloy cast irons, copper and its alloys – brass, bronze and cupronickel – aluminium and its alloys; Al–Cu precipitation strengthening treatment – titanium alloys, Mg-alloys, Ni-based super alloys – shape memory alloys – properties and applications – overview of materials standards.
UNIT IV	NON-METALLIC MATERIALS Polymers – types of polymers, commodity and engineering polymers – properties and applications of PE, PP, PS, PVC, PMMA, PET, PC, PA, ABS, PAI, PPO, PPS, PEEK, PTFE, thermoset polymers, urea and phenol formaldehydes – nylon, engineering ceramics – properties and applications of Al_2O_3 , SiC, Si_3N_4 , PSZ and SIALON – intermetallics – composites – matrix and reinforcement materials – applications of composites – nano composites.
UNIT V	MECHANICAL PROPERTIES AND DEFORMATION MECHANISMS Mechanisms of plastic deformation, slip and twinning – types of fracture – fracture mechanics – Griffith's theory – testing of materials under tension, compression and shear loads – hardness tests (Brinell, Vickers and Rockwell), micro and nano-hardness tests, impact test Izod and Charpy, fatigue and creep failure mechanisms.

Total: 45 periods (9 periods each)

Course Outcomes

At the end of the course, the students would be able to:

1. Explain alloys and phase diagrams, the Iron–Iron carbon diagram and steel classification.
2. Explain isothermal transformation, continuous cooling diagrams and different heat treatment processes.
3. Clarify the effect of alloying elements on ferrous and non-ferrous metals.
4. Summarize the properties and applications of non-metallic materials.
5. Explain the testing of mechanical properties.

Unit IV Learning Focus

For examination preparation, Unit IV should be studied with the following sequence:

1. **Polymer fundamentals:** polymer chain, repeat unit, thermoplastics, thermosets and elastomeric behaviour; commodity versus engineering polymers.
2. **Commodity polymers:** PE, PP, PS and PVC with their characteristic properties and daily/industrial applications.
3. **General engineering polymers:** PMMA, PET, PC, PA/nylon and ABS.
4. **High-performance engineering polymers:** PAI, PPO, PPS, PEEK and PTFE, with emphasis on heat, chemical, wear and dimensional performance.
5. **Thermosetting polymers:** urea-formaldehyde and phenol-formaldehyde resins.
6. **Engineering ceramics:** Al_2O_3 , SiC, Si_3N_4 , PSZ and SIALON – properties, limitations and applications.
7. **Intermetallics:** introductory concept, ordered crystal structure and engineering relevance.
8. **Composites:** matrix, reinforcement, interface, classification, fibre orientation and applications.
9. **Nanocomposites:** basic concept, nanoscale reinforcement, key benefits, limitations and example applications.

Engineering Context, Indian Relevance and Innovation Link

Why this unit matters: Modern mechanical systems increasingly combine metals with polymers, ceramics and composites to reduce mass, resist corrosion, improve wear performance, control heat/electricity and produce complex shapes economically.

Indian manufacturing relevance

- Commodity polymers are used in packaging, pipes, water-storage products, appliances, electrical insulation and consumer goods.
- Engineering plastics are used in automotive interiors, under-bonnet parts, gears, bearings, electrical connectors, pumps, medical devices and electronics housings.
- Alumina, SiC, Si_3N_4 , zirconia-based ceramics and SIALON are relevant to wear parts, seals, cutting tools, electrical insulation and high-temperature processing.
- GFRP and CFRP are increasingly used in wind-energy blades, transport, aerospace, sports equipment, pressure vessels and civil strengthening.
- Polymer and composite recycling, repair, joining and end-of-life design are growing engineering concerns alongside performance.

Start-up and project ideas linked with Unit IV

1. **Polymer selection assistant:** A simple tool that recommends a polymer from temperature, chemical exposure, transparency, impact and cost requirements.
2. **Plastic identification learning kit:** A safe classroom kit linking density, appearance, flexibility and recycling code to common polymer families.
3. **Composite lay-up demonstrator:** Small GFRP specimens with different fibre orientations to compare stiffness qualitatively.
4. **Ceramic application selector:** A database linking hardness, thermal shock, electrical insulation and wear resistance with Al_2O_3 , SiC, Si_3N_4 , PSZ and SIALON.
5. **AI-assisted material-selection exercise:** Students compare candidate polymer/composite materials, verify AI-generated claims against datasheets and explain the final engineering choice.

High-technology facts related to Unit IV

- High-performance thermoplastics such as PEEK can replace metals in selected hot, chemically aggressive and wear-related applications.
- Transformation-toughened zirconia uses a local phase transformation near a crack tip to improve fracture resistance.

- Composite properties can be deliberately directional: reinforcement is placed primarily along the load path.
- Nanocomposites use extremely small reinforcements and large interface area to modify mechanical, barrier, electrical or thermal behaviour at relatively low addition levels.

Text Books

1. Kenneth G. Budinski and Michael K. Budinski, *Engineering Materials*, Prentice Hall of India Private Limited, 9th Edition, 2018.
2. Sydney H. Avner, *Introduction to Physical Metallurgy*, McGraw Hill Book Company, 1994.

References

1. A. Alavudeen, N. Venkateshwaran, and J. T. Winowlin Jappes, *A Textbook of Engineering Materials and Metallurgy*, Laxmi Publications, 2006.
2. Amandeep Singh Wadhwa and Harvinder Singh Dhaliwal, *A Textbook of Engineering Material and Metallurgy*, University Sciences Press, 2008.
3. G. S. Upadhyay and Anish Upadhyay, *Materials Science and Engineering*, Viva Books Pvt. Ltd., New Delhi, 2020.
4. V. Raghavan, *Materials Science and Engineering*, Prentice Hall of India Pvt. Ltd., 6th Edition, 2019.
5. William D. Callister, *Materials Science and Engineering*, Wiley India Pvt. Ltd., 2nd Edition Reprint, 2019.
6. User-provided course reference: *ME3392 Engineering Materials and Metallurgy – Unit IV: Non-Metallic Materials*.

Source alignment note: The supplied Unit IV handout is used as the main source for commodity/engineering polymers, thermosetting resins, engineering ceramics and composite fundamentals. The syllabus topics **inter-metallics** and **nanocomposites** are not treated in comparable detail in the uploaded handout; those two topics are therefore clearly presented as introductory textbook-level completion for syllabus coverage.

1 Introduction to Unit IV

Non-metallic engineering materials are used when low density, corrosion resistance, electrical insulation, heat resistance, wear resistance, optical transparency or easy shaping is more important than the traditional advantages of metals.

The three major material groups in this unit are:

- **Polymers** – long-chain organic materials; generally light and easy to process.
- **Ceramics** – inorganic non-metallic materials; generally hard, heat resistant and brittle.
- **Composites** – combinations of two or more distinct constituents that work together to provide improved properties.

Structure + Processing → Properties → Application

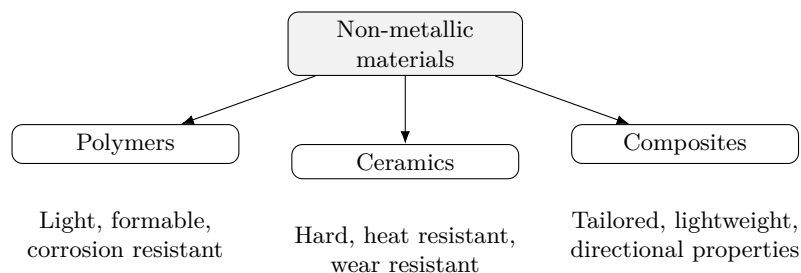


Figure 1: Main families considered in Unit IV.

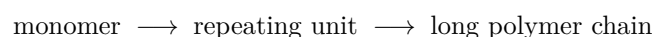
Exam point: For each material, connect its structure and property to a practical application.

2 Polymer Fundamentals and Classification

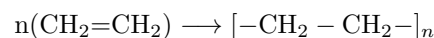
2.1 What Is a Polymer?

A polymer is a very large molecule made by joining many smaller repeating units called **monomers**. The chain nature of polymers explains many of their characteristic features such as low density, viscoelastic behaviour and easy shaping.

A simple representation is:



For polyethylene:



2.2 Classification by Thermal Behaviour

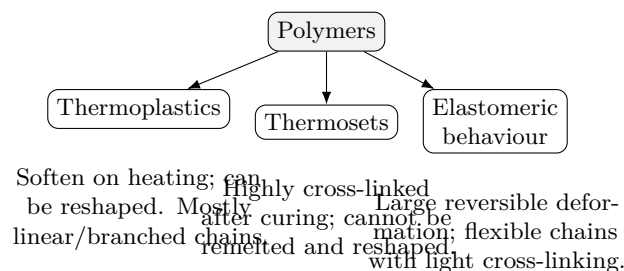


Figure 2: Basic classification of polymer behaviour.

Feature	Thermoplastics	Thermosets
Molecular structure	Linear/branched, limited cross-linking	Three-dimensional cross-linked network
Heating	Soften/melt on reheating	Do not remelt after curing
Processing	Injection moulding, extrusion, thermoforming	Compression moulding, casting, curing
Recycling by remelting	Generally more feasible	Not possible by simple remelting
Examples	PE, PP, PS, PVC, PC, nylon, PEEK	Phenol-formaldehyde, urea-formaldehyde

Table 2: Thermoplastics and thermosets – basic comparison.

2.3 Commodity and Engineering Polymers

The supplied course reference classifies high-volume polymers such as polyethylene, polypropylene and PVC as **commodity polymers**, while polymers with improved mechanical, thermal, dimensional and chemical performance are grouped as **engineering polymers**.

Feature	Commodity polymers	Engineering polymers
Primary goal	Low cost, high-volume use	Higher performance and reliability
Typical loading	Light to moderate	Moderate to demanding engineering service
Heat resistance	Generally lower	Generally higher
Dimensional stability	Moderate	Better controlled
Examples	PE, PP, PS, PVC	PC, PA, ABS, PPO, PPS, PEEK, PAI
Typical uses	Packaging, pipes, containers	Gears, housings, connectors, pumps, automotive components

Table 3: Commodity and engineering polymers.

2.4 Important Structure–Property Ideas

Polymer performance depends strongly on:

- molecular weight and chain length,
- linear, branched or cross-linked structure,
- crystallinity,
- chain stiffness and bulky side groups,
- glass transition temperature (T_g),
- melting temperature (T_m) for semicrystalline thermoplastics,
- fillers, plasticisers, fibres, stabilisers and flame retardants,
- service temperature, time and environment.

Remember: T_g is associated with a change from glassy/stiff to more rubber-like chain mobility in amorphous regions; T_m is the melting of crystalline regions.

3 Commodity Polymers

3.1 Polyethylene (PE)

Polyethylene is one of the most widely used polymers. Its repeat unit is simple, and changes in chain branching and molecular weight produce distinctly different grades.

Type	Structural feature	Typical property / use
LDPE	More branching	Flexible films, bags, squeeze bottles, wire insulation
HDPE	More linear, closer packing	Stronger/stiffer bottles, drums, pipes, containers
LLDPE	Linear chain with short branches	Tough flexible films
UHMWPE	Extremely high molecular weight	Very high wear resistance; liners, sliding parts, high-strength fibres
XLPE	Cross-linked PE network	Improved stress-crack and thermal performance; cable insulation, tanks

Table 4: Common forms of polyethylene.

Important properties: low density, good chemical resistance, good electrical insulation, low water absorption and easy processing.

Limitations: relatively low service temperature, creep under sustained load and lower stiffness than many engineering plastics.

Applications: Packaging films, bottles, pipes, tanks, cable insulation, toys, low-friction liners and wear strips.

3.2 Polypropylene (PP)

PP is a versatile commodity thermoplastic used both as a plastic and as a fibre. The reference source highlights its low water absorption, useful heat resistance relative to PE and good stress-crack resistance.

Properties

- low density and good chemical resistance,
- higher stiffness and heat resistance than many PE grades,
- low moisture absorption,
- good fatigue resistance for repeated flexing,
- can become brittle at low temperature in some grades.

Applications: Living hinges, reusable food containers, automotive trim, appliance parts, laboratory ware, carpets, ropes, fibres and packaging.

3.3 Polystyrene (PS)

PS is an amorphous thermoplastic that is normally rigid, economical and easy to mould. General-purpose PS is relatively brittle, while modifications improve impact behaviour.

Forms and applications

- **General-purpose PS:** transparent rigid items, model parts, disposable articles.
- **HIPS (high-impact PS):** rubber-toughened PS for toys and product housings.
- **EPS (expanded PS):** low-density foam for packaging and thermal insulation.

Applications: Protective packaging, insulation, disposable cups/trays, appliance housings, models and display components.

3.4 Polyvinyl Chloride (PVC)

PVC is an amorphous thermoplastic whose properties can be changed greatly using stabilisers, plasticisers, fillers and impact modifiers.

Type	Main character	Examples
Rigid PVC	Stiff, corrosion resistant, economical	Pipes, fittings, profiles, window sections, sheets
Flexible PVC	Plasticised, flexible and insulating	Cable insulation, flooring, flexible tubing, synthetic leather

Table 5: Rigid and flexible PVC.

Advantages: corrosion resistance, electrical insulation, good weatherability in suitable grades and low cost.

Limitations: thermal stability requires appropriate formulation; mechanical properties depend strongly on plasticiser/additive content.

Applications: Water and drainage pipes, electrical conduit, cable insulation, flooring, profiles, medical tubing and coated fabrics.

4 General Engineering Thermoplastics

4.1 PMMA – Poly(methyl methacrylate)

PMMA is a transparent amorphous thermoplastic known for excellent optical clarity and weather resistance.

Properties

- excellent transparency and optical quality,
- good UV and weather resistance,
- hard and stiff surface,
- lower impact toughness than polycarbonate,
- relatively notch-sensitive and scratch-sensitive.

Applications: Transparent guards, displays, signs, lenses, vehicle tail-light covers, aquarium windows, aircraft glazing and optical components.

Remember: PMMA is selected when optical clarity and weatherability are more important than very high impact resistance.

4.2 PET – Polyethylene Terephthalate

PET is a thermoplastic polyester with good strength, stiffness, dimensional stability, low water absorption and useful gas-barrier behaviour. The reference notes its use in oriented films, fibres and bottles.

Properties

- good strength and stiffness,
- low moisture absorption,
- good gas-barrier properties,
- good chemical resistance except to strong alkaline hydrolysis,
- can be oriented to improve strength, toughness and barrier performance.

Applications: Carbonated-drink bottles, food packaging, polyester fibres, electrical films, recording/insulating films and engineering components in reinforced grades.

4.3 Polycarbonate (PC)

PC is an amorphous engineering thermoplastic noted in the reference source for its unusual combination of transparency, toughness, dimensional stability and heat resistance.

Properties

- very high impact strength,
- optical transparency,
- good dimensional stability,
- good electrical insulation,
- useful heat resistance,
- lower chemical/UV resistance than some alternatives unless stabilised.

Applications: Safety glazing, face shields, helmets, headlamp lenses, electrical housings, machine guards, medical components and electronic-device housings.

Requirement	PMMA	PC
Optical clarity	Excellent	Excellent
Impact resistance	Moderate / brittle compared with PC	Very high
Scratch resistance	Better than PC in many grades	Lower; coatings may be used
Typical choice	Displays, signs, lenses	Safety glazing, helmets, impact-resistant housings

Table 6: Simple PMMA–PC selection comparison.

4.4 Polyamides (PA) / Nylon

Polyamides contain amide groups in the main chain and are widely known as nylons. Common engineering grades include PA6 and PA66.

Properties

- high strength and toughness,
- good fatigue and abrasion resistance,
- low coefficient of friction and useful self-lubricating behaviour,
- good resistance to oils, fuels and many engineering fluids,
- moisture absorption affects dimensions and mechanical properties.

Applications: Gears, bearings, rollers, fan blades, cable ties, power-tool housings, automotive intake components, textile fibres and fishing line.

Exam point: In engineering design, nylon moisture absorption must be considered because it can change dimensions, stiffness and toughness.

4.5 ABS – Acrylonitrile Butadiene Styrene

ABS combines a styrene-acrylonitrile matrix with a rubbery butadiene phase. This gives a useful balance of toughness, rigidity and surface quality.

Properties

- high impact resistance and toughness,
- good surface finish and dimensional stability,
- useful stiffness and chemical resistance,
- easy injection moulding and thermoforming,
- generally poor weathering without protection and limited flame resistance in ordinary grades.

Applications: Automotive interior trim, instrument panels, helmets, appliance housings, luggage shells, pipes and fittings, communication-equipment housings, and electroplated decorative parts.

5 High-Performance Engineering Polymers

5.1 PAI – Polyamide-Imide

PAI is described in the source as a high-performance amorphous engineering thermoplastic combining high strength, heat capability, wear resistance and broad chemical resistance.

Properties

- very high strength among unreinforced thermoplastics,
- excellent elevated-temperature mechanical properties,
- good wear and creep resistance,
- low smoke and good radiation resistance,
- expensive and processing conditions are demanding.

Applications: Valves, pumps, bearings, gears, electrical connectors, jet-engine components, bearing cages, seals and high-temperature mechanical parts.

5.2 PPO – Polyphenylene Oxide

PPO has high strength, heat resistance and dimensional stability. The reference source notes that commercial PPO is commonly blended with polystyrene to improve processability and reduce cost.

Properties

- good heat-deflection resistance,
- excellent dimensional stability and low moisture absorption,
- good electrical properties,
- useful hydrolysis resistance,
- poor resistance to several aromatic/chlorinated solvents,
- often used as modified PPO/PPO-PS blends.

Applications: Electrical housings, fuse boxes, appliance frames, automotive fascia panels, pump housings, computer housings and structural-foam parts.

5.3 PPS – Polyphenylene Sulphide

PPS is a crystalline high-performance engineering thermoplastic with high heat, chemical and flame resistance. It is commonly reinforced with glass fibres or mineral fillers.

Properties

- high-temperature capability,
- excellent chemical resistance,
- good dimensional stability,
- good radiation resistance,
- self-extinguishing behaviour in many grades,
- higher cost than commodity plastics.

Applications: Electrical sockets/connectors, chemical-pump parts, coatings, microwave/appliance components and under-bonnet automotive parts.

5.4 PEEK – Polyether Ether Ketone

PEEK is a semicrystalline high-performance thermoplastic used where thermal, chemical, fatigue and wear properties are critical.

Properties

- high strength and stiffness over a wide temperature range,
- excellent chemical resistance,
- good fatigue, wear and hydrolysis resistance,
- inherently good flame resistance with low smoke,
- can be reinforced with glass or carbon fibres,
- very expensive and requires high processing temperature.

Applications: High-temperature wire insulation, pump and valve parts, seals, bearings, chemical-processing equipment, oilfield components, medical/aerospace parts and metal-replacement components.

5.5 PTFE – Polytetrafluoroethylene

PTFE is a fluoropolymer known for exceptional chemical inertness, very low friction and excellent electrical insulation.

Properties

- extremely low coefficient of friction,
- excellent chemical and solvent resistance,

- excellent electrical insulation,
- very low water absorption and non-wetting surface,
- useful over a very wide temperature range,
- relatively low stiffness/strength and difficult conventional melt processing.

Applications: Non-stick coatings, chemical vessel linings, gaskets, seals, piston rings, bearings, low-friction guides and high-grade electrical insulation.

Polymer	Main property emphasis	Typical limitation	Representative application
PAI	Strength + heat + wear	High cost/processing demand	Bearings, seals, jet-engine parts
PPO	Dimensional stability + electrical	Solvent resistance	Appliance/electrical housings
PPS	Heat + chemicals + flame resistance	Cost/brittleness in some grades	Connectors, pump parts
PEEK	Heat + fatigue + chemicals	Very high cost	Valves, seals, aerospace/oilfield parts
PTFE	Lowest friction + chemical inertness	Low stiffness, difficult processing	Gaskets, linings, bearings

Table 7: High-performance polymers – property/application emphasis.

6 Thermosetting Polymers

6.1 General Behaviour

Thermosetting polymers form a cross-linked network during curing. After curing, they do not soften and flow like thermoplastics when reheated.

General advantages: heat resistance, dimensional stability, chemical resistance and good electrical insulation.

General limitations: comparatively brittle behaviour and inability to be remelted for simple reshaping/recycling.

6.2 Urea–Formaldehyde (UF)

UF is a thermosetting resin produced from urea and formaldehyde by condensation reaction.

Important characteristics

- good adhesive behaviour for wood products,
- economical and reactive curing system,
- light-coloured appearance,
- used mainly in interior wood products and binders.

Applications: Particleboard, plywood, MDF, laminated wood products, decorative laminates, foundry cores, paper/textile treatment and wood adhesives.

6.3 Phenol–Formaldehyde (PF) / Phenolic Resin

PF is a thermosetting resin formed by condensation polymerisation of phenol and formaldehyde. The traditional name Bakelite is associated with phenolic moulding compounds.

Properties

- useful heat resistance,
- good dimensional stability,
- good electrical insulation in suitable conditions,
- good resistance to many chemicals,
- hard and comparatively brittle.

Applications: Electrical plugs/switch components, heat-resistant handles/knobs, laminates, brake/friction-related binders, adhesives, coatings and composite matrices.

Feature	UF	PF
Typical colour	Light/white	Dark brown/black
Main use emphasis	Wood adhesives and panels	Heat-resistant mouldings, laminates, electrical parts
Heat resistance	Moderate	Generally better
Typical product	Particleboard adhesive	Cooker handle / electrical component

Table 8: Simple comparison of urea-formaldehyde and phenol-formaldehyde resins.

7 Engineering Ceramics

7.1 Basic Characteristics

Ceramics are inorganic non-metallic materials whose bonding is mainly ionic and/or covalent. Engineering ceramics are designed for demanding mechanical, thermal, electrical or chemical service.

Typical advantages

- very high hardness and wear resistance,
- high compressive strength,
- high-temperature stability,
- corrosion and oxidation resistance,
- electrical insulation in many ceramic systems,
- low density compared with many metals.

Typical limitations

- low tensile ductility,
- brittle fracture and flaw sensitivity,
- difficult machining and joining,
- comparatively high processing cost for precision components.

Ceramics are often selected because they survive heat, wear, corrosion or electrical conditions that are difficult for ordinary metals and polymers.

7.2 Aluminium Oxide – Al_2O_3 (Alumina)

Alumina is one of the most widely used engineering ceramics because it provides a useful combination of hardness, electrical insulation, temperature capability and cost.

Properties

- high hardness and wear resistance,
- high stiffness and compressive strength,
- excellent dielectric behaviour,
- good chemical resistance,
- good high-temperature stability.

Applications: Electrical insulators, electronic substrates, seal rings, wear pads, grinding media, furnace tubes, thread/wire guides and armour components.

7.3 Silicon Carbide – SiC

SiC is a very hard, low-density ceramic with excellent wear, corrosion and thermal-shock resistance.

Properties

- extremely high hardness,
- low density,

- high thermal conductivity,
- low thermal expansion,
- excellent wear and corrosion resistance,
- strong thermal-shock resistance.

Applications: Mechanical seals, pump bearings, blast nozzles, heat-exchanger tubes, armour, semiconductor handling components and chemical-process valve parts.

7.4 Silicon Nitride – Si_3N_4

Silicon nitride combines high strength, hardness, wear resistance and good thermal shock behaviour.

Properties

- high strength over a wide temperature range,
- better fracture toughness than many conventional ceramics,
- high hardness and wear resistance,
- good thermal shock resistance,
- good chemical resistance.

Applications: Bearing balls/rollers, cutting tools, turbocharger rotors, engine valves and wear parts, forming rolls, shafts and high-wear precision components.

7.5 Partially Stabilized Zirconia – PSZ

Pure zirconia undergoes crystal-structure changes during cooling. Suitable stabilising oxides such as MgO , CaO or Y_2O_3 can control these transformations. In PSZ, retained tetragonal regions can transform near a crack and improve toughness.

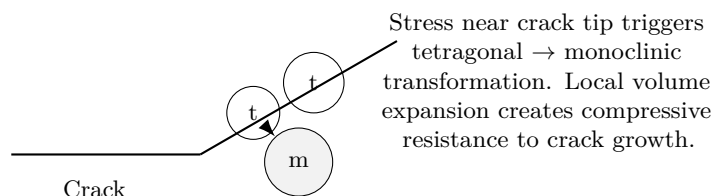


Figure 3: Simplified transformation-toughening concept in partially stabilised zirconia.

Properties: high fracture toughness and hardness, wear resistance, low thermal conductivity and chemical stability.

Applications: Valve balls/seats, pump seals, wire guides, wear parts, cutting/wear components and oxygen-sensor-related zirconia systems.

7.6 SIALON

SIALON is a family of silicon-aluminium-oxygen-nitrogen ceramics derived from silicon nitride chemistry. The supplied source emphasises its thermal-shock resistance and resistance to wetting/corrosion by non-ferrous molten metals.

Properties

- high strength and useful fracture toughness,
- excellent thermal shock resistance,
- low thermal expansion,
- good oxidation and wear resistance,
- resistance to attack/wetting by many non-ferrous molten metals.

Applications: Cutting tools, aluminium die-casting feed tubes, thermocouple protection tubes, metal-forming tools, welding fixtures and wear components.

7.7 Comparison of Engineering Ceramics

Ceramic	Main strength	Limitation / caution	Typical application
Al ₂ O ₃	Hard, insulating, economical	Brittle / moderate thermal shock	Insulators, substrates, wear pads
SiC	Hard + thermal conductivity + corrosion	Difficult precision machining	Seals, nozzles, heat exchangers
Si ₃ N ₄	Strength + toughness + thermal shock	Cost	Bearings, engine/turbo parts
PSZ	High fracture toughness + wear	Properties depend on stabilisation/service	Valves, seals, wear parts
SIALON	Thermal shock + molten-metal resistance	Specialised/costly	Cutting tools, casting hardware

Table 9: Engineering ceramics – property/application comparison.

8 Intermetallics

Textbook completion for syllabus coverage: The uploaded Unit IV reference does not develop intermetallics; the following is an introductory syllabus-level treatment based on standard materials-science terminology.

An **intermetallic compound** is a solid phase formed between two or more metallic elements with an ordered crystal structure and a composition near a particular stoichiometric ratio. Intermetallics differ from ordinary random solid solutions because atoms preferentially occupy specific lattice sites.

Typical characteristics

- high specific strength and good high-temperature strength in selected systems,
- good oxidation/corrosion resistance in selected systems,
- ordered crystal structures and strong bonding,
- lower room-temperature ductility and difficult processing in many systems.

Examples and relevance

- Ni₃Al – important strengthening phase in nickel-based superalloys,
- NiAl – high-temperature and oxidation-resistant intermetallic,
- TiAl (titanium aluminide) – low density and high-temperature capability for selected aero-engine/automotive applications,
- FeAl – oxidation/corrosion-resistant iron aluminide family.

Applications: High-temperature aerospace components, selected turbine/engine parts, coatings and strengthening precipitates within advanced alloys.

9 Composite Materials

9.1 Definition and Basic Concept

A composite is a multiphase material containing at least two distinct constituents that remain separated by an interface and work together to provide a useful combination of properties.

Most engineering composites contain:

- a continuous **matrix**, and
- a stronger/stiffer **reinforcement** such as fibres, particles or whiskers.

Matrix holds and protects; reinforcement carries much of the load; the interface transfers load between them.

9.2 Functions of Matrix and Reinforcement

Constituent	Main functions	Examples
Matrix	Gives shape, transfers load, protects reinforcement, resists environment, supports fibres in compression	Epoxy, polyester, Al, Mg, ceramic matrices
Reinforcement	Provides strength, stiffness, wear or functional enhancement	Glass fibre, carbon fibre, aramid fibre, SiC particles/whiskers
Interface	Transfers stress and controls damage/debonding	Fibre–resin bond, particle–matrix bond

Table 10: Functions of composite constituents.

9.3 Classification by Matrix

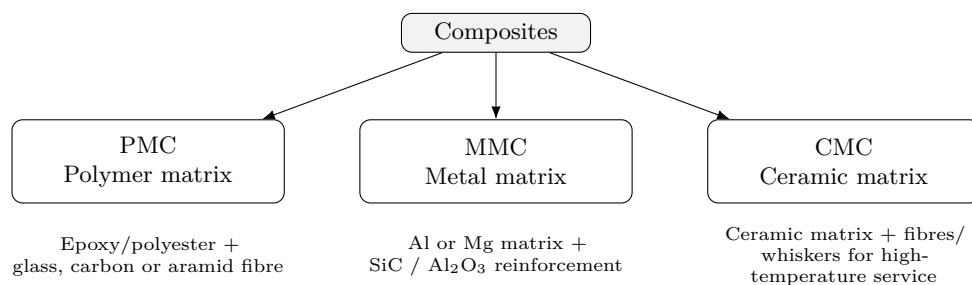


Figure 4: Classification of composites based on matrix material.

9.4 Classification by Reinforcement Form

- **Fibre reinforced:** continuous or short fibres.
- **Particulate:** particles or flakes distributed in matrix.
- **Laminar/laminated:** stacked layers with selected orientations.
- **Hybrid:** more than one reinforcement type or fibre family.

9.5 Common Matrix Materials

Polymer matrices: epoxy, polyester and other thermosets/thermoplastics. They are easy to shape and corrosion resistant but temperature capability is limited compared with metals/ceramics.

Metal matrices: aluminium and magnesium are common lightweight matrices; higher temperature and transverse strength than PMCs, but processing is more difficult.

Ceramic matrices: retain properties at high temperature and are reinforced to improve damage tolerance relative to monolithic ceramics.

9.6 Common Reinforcements

- **Glass fibre:** economical, good tensile strength, corrosion resistant, electrically insulating.
- **Carbon fibre:** very high specific stiffness/strength, low thermal expansion, electrically conductive, high cost.
- **Aramid fibre:** low density, high tensile strength and impact/energy absorption, weaker in compression than carbon fibre.
- **Ceramic particles/whiskers:** SiC, Al₂O₃ and similar reinforcements improve stiffness, wear and temperature capability.

9.7 Fibre Orientation and Anisotropy

Fibres are strongest and stiffest mainly along their length. Therefore composite properties depend strongly on orientation.

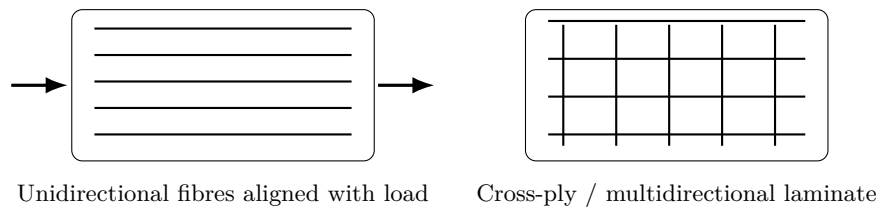


Figure 5: Fibre orientation is selected according to the direction of applied load.

Remember: Composite design must consider both magnitude and direction of load. This directional behaviour is called anisotropy.

9.8 Fibre Volume Fraction and Manufacturing Quality

The source notes that composite properties depend on fibre properties, resin properties, fibre-to-resin ratio and fibre geometry/orientation. Higher fibre volume fraction generally increases stiffness and strength up to practical processing limits, provided fibres are properly wetted and voids are controlled.

- Too little reinforcement: matrix dominates, lower stiffness/strength.
- Optimum reinforcement: efficient load carrying and good wetting.
- Excessive/poorly packed reinforcement: dry regions and voids may reduce quality.

9.9 Behaviour Under Basic Loads

Load	Important composite response
Tension	Fibres carry a large share of load when aligned with the tensile direction.
Compression	Matrix supports fibres and helps prevent fibre buckling.
Shear	Matrix and interface transfer stress between adjacent fibres/layers.
Flexure	Combination of tension, compression and shear through laminate thickness.

Table 11: Role of constituents under common mechanical loads.

9.10 Applications of Composites

Sector	Example components	Why composites?
Aerospace	Wing/tail panels, fairings, interior structures	High specific stiffness/strength, corrosion resistance
Automotive	Body panels, leaf springs, drive shafts, mono-coques	Weight reduction, parts integration
Wind energy	Wind-turbine blades	Long lightweight stiff structures
Marine	Boat hulls, decks, masts	Corrosion resistance + low mass
Sports	Bicycles, rackets, helmets, golf shafts	Low weight + tailored stiffness
Civil	FRP strengthening wraps, rebars, bridge panels	Corrosion resistance, rapid installation
Pressure systems	Composite cylinders/tanks	High strength-to-weight ratio
Electrical	GFRP insulators, circuit laminates	Electrical insulation + structural support

Table 12: Practical applications of composite materials.

9.11 Advantages and Limitations

Advantages

- high specific strength and stiffness,
- good corrosion resistance,
- properties can be tailored by fibre orientation,
- good fatigue behaviour in many applications,
- ability to mould large integrated shapes.

Limitations

- anisotropy complicates design,
- damage may occur internally as delamination or matrix cracking,
- repair and inspection require special methods,
- processing quality strongly affects performance,
- recycling and end-of-life treatment can be difficult.

10 Nanocomposites

Textbook completion for syllabus coverage: The uploaded Unit IV reference discusses conventional fibre and particulate composites but does not develop nanocomposites. The following is an introductory textbook-level completion for the prescribed syllabus.

10.1 Definition and Basic Principle

A nanocomposite is a composite in which at least one reinforcement dimension is in the nanometre range. Typical reinforcements include nanoparticles, nanoclays, carbon nanotubes (CNTs), graphene/graphene nanoplatelets and nanoscale ceramic particles.

The main reason nanoscale additions can be effective is their extremely large surface area relative to volume:

Small particle size $\Rightarrow$ large interface area $\Rightarrow$ strong influence on matrix behaviour

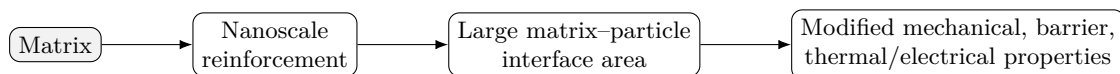


Figure 6: Basic nanocomposite concept.

10.2 Common Types

- **Polymer nanocomposites:** polymer + nanoclay, SiO₂, CNT, graphene, Al₂O₃ nanoparticles.
- **Metal matrix nanocomposites:** Al/Mg and other metals containing nanoscale ceramic or carbon reinforcement.
- **Ceramic nanocomposites:** ceramic matrices modified by nanoscale second phases for toughness, wear or functional response.

10.3 Possible Property Improvements

Depending on dispersion, interface and reinforcement type, nanocomposites may provide:

- increased stiffness/strength,
- improved wear and scratch resistance,
- improved gas/moisture barrier behaviour,
- electrical conductivity or antistatic behaviour,
- improved thermal conductivity or flame resistance,
- improved multifunctionality at relatively low reinforcement fraction.

10.4 Major Challenges

- nanoparticles tend to agglomerate,
- uniform dispersion is difficult,
- interface bonding controls effectiveness,
- viscosity/processing may become difficult,
- health, handling and environmental considerations are important for free nanoparticles,

- cost and reproducibility can limit industrial adoption.

Applications: Barrier packaging, conductive/antistatic polymers, wear-resistant coatings, lightweight structural parts, sensors, electromagnetic shielding and multifunctional composite components.

11 Material Selection – Quick Engineering Guide

Design requirement	Likely material family	Example
Very low-cost flexible packaging	PE / PP	Films, bags, food containers
Rigid pipe with corrosion resistance	PVC / HDPE	Water/drainage pipe
Optical transparency + weathering	PMMA	Signs, covers, glazing
Transparent + very high impact	PC	Helmets, machine guards
Low-friction gear/bearing	Nylon / PTFE / PEEK	Gears, bushings, seals
High heat + chemical resistance polymer	PPS / PEEK / PAI	Pumps, connectors, hot components
Electrical insulation + hardness	Alumina	Insulators, substrates
Wear + thermal-shock ceramic	SiC / Si ₃ N ₄ / SiALON	Seals, cutting tools, turbo components
High ceramic toughness	PSZ	Valve seats, wear parts
Very high specific stiffness/strength	CFRP/GFRP	Aerospace, wind blades, sports
Tailored multifunctionality with nano-addition	Nanocomposite	Barrier, conductive or wear-modified parts

Table 13: Simple property-to-material selection guide for Unit IV.

12 Important Exam-Oriented Points

12.1 Very Short Answer Points

1. Thermoplastics soften on heating; thermosets form cross-linked networks and do not remelt after curing.
2. Commodity polymers are used in high volume and are generally lower cost; engineering polymers provide higher performance.
3. PE families include LDPE, HDPE, LLDPE, UHMWPE and XLPE.
4. PP is low-density, chemically resistant and useful for repeated flexing/living hinges.
5. PS is economical and rigid; HIPS improves impact behaviour and EPS provides lightweight insulation.
6. PVC may be rigid or plasticised/flexible.
7. PMMA provides excellent optical clarity and weather resistance.
8. PET provides strength, dimensional stability and good gas-barrier properties.
9. PC combines high impact strength with transparency.
10. Polyamides/nylons provide strength, abrasion resistance and low friction but absorb moisture.
11. ABS provides toughness, rigidity and good surface finish.
12. PAI is a high-strength, high-temperature engineering thermoplastic.
13. PPO provides dimensional stability and low moisture absorption; commercial grades are often blends.
14. PPS provides high heat and chemical resistance.
15. PEEK combines high-temperature, chemical, fatigue and wear resistance.
16. PTFE has extremely low friction and exceptional chemical resistance.
17. UF is widely used as a wood-panel adhesive/binder.
18. PF is a heat-resistant thermosetting resin used in mouldings, laminates and adhesives.
19. Alumina is a hard, wear-resistant electrical insulating ceramic.
20. SiC combines high hardness, thermal conductivity and thermal-shock resistance.
21. Si₃N₄ combines strength, wear resistance and good thermal-shock behaviour.
22. PSZ uses controlled zirconia phase transformation to obtain high toughness.

23. SIALON is suited to cutting tools and non-ferrous molten-metal contact.
24. Intermetallics are ordered phases between metallic elements; examples include Ni_3Al and TiAl .
25. Matrix holds/protects reinforcement; reinforcement provides strength/stiffness; the interface transfers load.
26. PMC, MMC and CMC denote polymer-, metal- and ceramic-matrix composites.
27. Fibre orientation causes anisotropic composite behaviour.
28. Nanocomposites use nanoscale reinforcement and very large interface area to modify properties.

12.2 Important Two-Mark Questions

1. Define polymer and monomer.
2. Differentiate thermoplastic and thermosetting polymers.
3. Differentiate commodity and engineering polymers with examples.
4. List the main types of polyethylene.
5. State two important properties and applications of polypropylene.
6. What is HIPS? What is EPS?
7. Differentiate rigid and flexible PVC.
8. Why is PMMA used for transparent applications?
9. State two properties and applications of PET.
10. Why is polycarbonate used for safety glazing?
11. What are polyamides? Give two applications of nylon.
12. State the main constituents and properties of ABS.
13. State two properties of PAI.
14. Why is PPO commonly modified/blended?
15. State two properties and uses of PPS.
16. Why is PEEK called a high-performance engineering polymer?
17. Why is PTFE suitable for bearings and chemical linings?
18. Define urea-formaldehyde and phenol-formaldehyde resins.
19. State two properties and applications of alumina.
20. Why is SiC useful for mechanical seals?
21. State two properties and applications of Si_3N_4 .
22. What is PSZ? What is transformation toughening?
23. State two properties and applications of SIALON.
24. Define intermetallic compound with one example.
25. Define composite and identify matrix and reinforcement.
26. Classify composites according to matrix and reinforcement form.
27. What is anisotropy in fibre-reinforced composites?
28. Define nanocomposite and list two nanoscale reinforcements.

12.3 Important Five-Mark Questions

1. Classify polymers and compare thermoplastics with thermosets.
2. Explain commodity and engineering polymers with suitable examples.
3. Explain properties and applications of PE, PP, PS and PVC.
4. Explain properties and applications of PMMA, PET and PC.

5. Explain properties and applications of PA/nylon and ABS.
6. Explain properties and applications of PAI, PPO and PPS.
7. Explain properties and applications of PEEK and PTFE.
8. Explain urea-formaldehyde and phenol-formaldehyde thermosetting resins.
9. Explain properties and applications of Al_2O_3 and SiC.
10. Explain properties and applications of Si_3N_4 , PSZ and SIALON.
11. Write a short note on intermetallics with examples and applications.
12. Explain matrix, reinforcement and interface in composites.
13. Classify composites into PMC, MMC and CMC with examples.
14. Explain the influence of fibre orientation and fibre volume fraction on composite properties.
15. Explain engineering applications of composites in aerospace, automotive, wind and civil sectors.
16. Explain nanocomposites, common reinforcements, advantages and limitations.

12.4 Important Long-Answer Questions

1. Classify polymers and discuss commodity and engineering polymers with suitable examples.
2. Explain properties and applications of PE, PP, PS, PVC, PMMA, PET, PC, PA/nylon and ABS.
3. Explain high-performance engineering polymers PAI, PPO, PPS, PEEK and PTFE with applications.
4. Explain thermosetting polymers with special reference to urea-formaldehyde and phenol-formaldehyde resins.
5. Explain the properties and applications of Al_2O_3 , SiC, Si_3N_4 , PSZ and SIALON engineering ceramics.
6. Explain intermetallics and discuss their important features and engineering relevance.
7. Define composites and explain matrix/reinforcement materials, classification, anisotropy and applications.
8. Explain nanocomposites with classification, strengthening concept, advantages, limitations and applications.

13 Unit Summary

- Polymers are long-chain materials whose properties depend on chain structure, crystallinity, cross-linking, temperature and additives.
- Thermoplastics can be softened and reshaped; thermosets form permanent cross-linked networks after curing.
- PE, PP, PS and PVC are major commodity polymers used in high-volume applications.
- PMMA is selected for optical clarity; PET for bottles/films/fibres; PC for impact-resistant transparent parts; nylon for wear/gear applications; ABS for tough housings and automotive trim.
- PAI, PPO, PPS, PEEK and PTFE are engineering/high-performance polymers selected for heat, chemicals, wear, dimensional stability or low friction.
- UF and PF are important thermosetting resins used in wood products, mouldings, laminates and adhesives.
- Engineering ceramics provide hardness, wear resistance, corrosion resistance and high-temperature capability, but are generally brittle and flaw sensitive.
- Alumina is a versatile electrical/wear ceramic; SiC excels in wear, corrosion and thermal shock; Si_3N_4 combines strength and thermal shock resistance; PSZ offers transformation toughening; SIALON is valuable for cutting and molten-metal contact.
- Intermetallics are ordered metallic compounds such as Ni_3Al and TiAl ; selected systems offer high-temperature capability but may have limited ductility.
- A composite combines matrix and reinforcement. Matrix provides shape/protection/load transfer; reinforcement provides high strength/stiffness.
- Composite properties depend strongly on fibre type, volume fraction, interface, manufacturing quality and orientation.

- PMCs dominate lightweight structures; MMCs and CMCs extend composite use to higher temperature and wear conditions.
- Nanocomposites use nanoscale reinforcements and large interface area to modify mechanical, barrier, thermal, electrical or wear behaviour.

14 Glossary of Important Terminologies

The following short definitions are useful for quick revision and two-mark examination answers.

Term	Short definition / exam point
Polymer	Large molecule made from many repeating molecular units.
Monomer	Small molecule that reacts to form a polymer chain.
Repeat unit	Structural unit repeated along a polymer chain.
Thermoplastic	Polymer that softens on heating and can generally be reshaped repeatedly.
Thermoset	Polymer forming a permanently cross-linked network during curing; does not remelt.
Elastomer	Polymer showing large recoverable elastic deformation.
Commodity polymer	High-volume, generally lower-cost polymer used mainly in less demanding applications.
Engineering polymer	Polymer designed for improved mechanical, thermal, dimensional or chemical performance.
Glass transition temperature (T_g)	Temperature range where amorphous polymer regions change from glassy/stiff to more mobile/rubbery behaviour.
Melting temperature (T_m)	Temperature at which crystalline regions of a semicrystalline thermoplastic melt.
Crystallinity	Fraction/order of polymer chains arranged in crystalline regions.
Plasticiser	Additive used to increase polymer flexibility and lower effective stiffness/softening behaviour.
LDPE	Low-density polyethylene with significant chain branching and high flexibility.
HDPE	High-density polyethylene with more linear chains and greater stiffness/strength.
UHMWPE	Ultra-high molecular weight polyethylene with excellent wear and impact performance.
XLPE	Cross-linked polyethylene with improved thermal/stress-crack resistance.
HIPS	High-impact polystyrene, usually rubber-toughened PS.
EPS	Expanded polystyrene foam used for insulation and packaging.
PMMA	Transparent acrylic thermoplastic with excellent optical clarity and weather resistance.
PET	Thermoplastic polyester used in bottles, films and fibres.
PC	Polycarbonate; transparent, tough engineering thermoplastic with high impact resistance.
PA / Nylon	Polyamide thermoplastics with good strength, abrasion resistance and low friction; moisture-sensitive.
ABS	Acrylonitrile-butadiene-styrene; tough engineering thermoplastic used for housings and automotive trim.
PAI	Polyamide-imide; high-strength, high-temperature engineering thermoplastic.
PPO	Polyphenylene oxide; heat-resistant, dimensionally stable polymer often used as modified blends.
PPS	Polyphenylene sulphide; heat- and chemical-resistant engineering thermoplastic.
PEEK	Polyether ether ketone; high-performance semicrystalline polymer with excellent heat/chemical/wear resistance.
PTFE	Polytetrafluoroethylene; fluoropolymer with extremely low friction and high chemical resistance.
Urea-formaldehyde	Thermosetting amino resin widely used as wood adhesive/binder.
Phenol-formaldehyde	Phenolic thermosetting resin used in heat-resistant mouldings, laminates and adhesives.
Engineering ceramic	Advanced inorganic non-metallic material engineered for structural/thermal/electrical service.
Alumina	Al_2O_3 ceramic; hard, wear resistant and electrically insulating.

Silicon carbide	SiC ceramic with high hardness, thermal conductivity and corrosion/thermal-shock resistance.
Silicon nitride	Si ₃ N ₄ ceramic combining strength, wear resistance and good thermal-shock performance.
PSZ	Partially stabilized zirconia; toughened zirconia containing controlled metastable tetragonal phase.
Transformation toughening	Crack-resistance mechanism using stress-induced phase transformation and local volume change.
SIALON	Silicon-aluminium-oxygen-nitrogen ceramic family derived from silicon nitride.
Intermetallic compound	Ordered solid phase between metallic elements, often near a fixed stoichiometric composition.
Composite	Material made of two or more distinct constituents acting together to provide improved properties.
Matrix	Continuous phase of a composite that gives shape, protects reinforcement and transfers load.
Reinforcement	Fibre, particle or other phase added primarily to improve stiffness, strength, wear or functional properties.
Interface	Boundary between matrix and reinforcement through which load and damage are transferred.
PMC	Polymer matrix composite.
MMC	Metal matrix composite.
CMC	Ceramic matrix composite.
GFRP	Glass-fibre-reinforced polymer.
CFRP	Carbon-fibre-reinforced polymer.
Fibre volume fraction	Volume of fibres divided by total composite volume; strongly affects composite properties.
Anisotropy	Direction-dependent material properties.
Delamination	Separation between layers of a laminate composite.
Nanocomposite	Composite containing a reinforcement with at least one dimension in the nanoscale range.
Agglomeration	Clustering of small particles, reducing uniformity and effective reinforcement.
